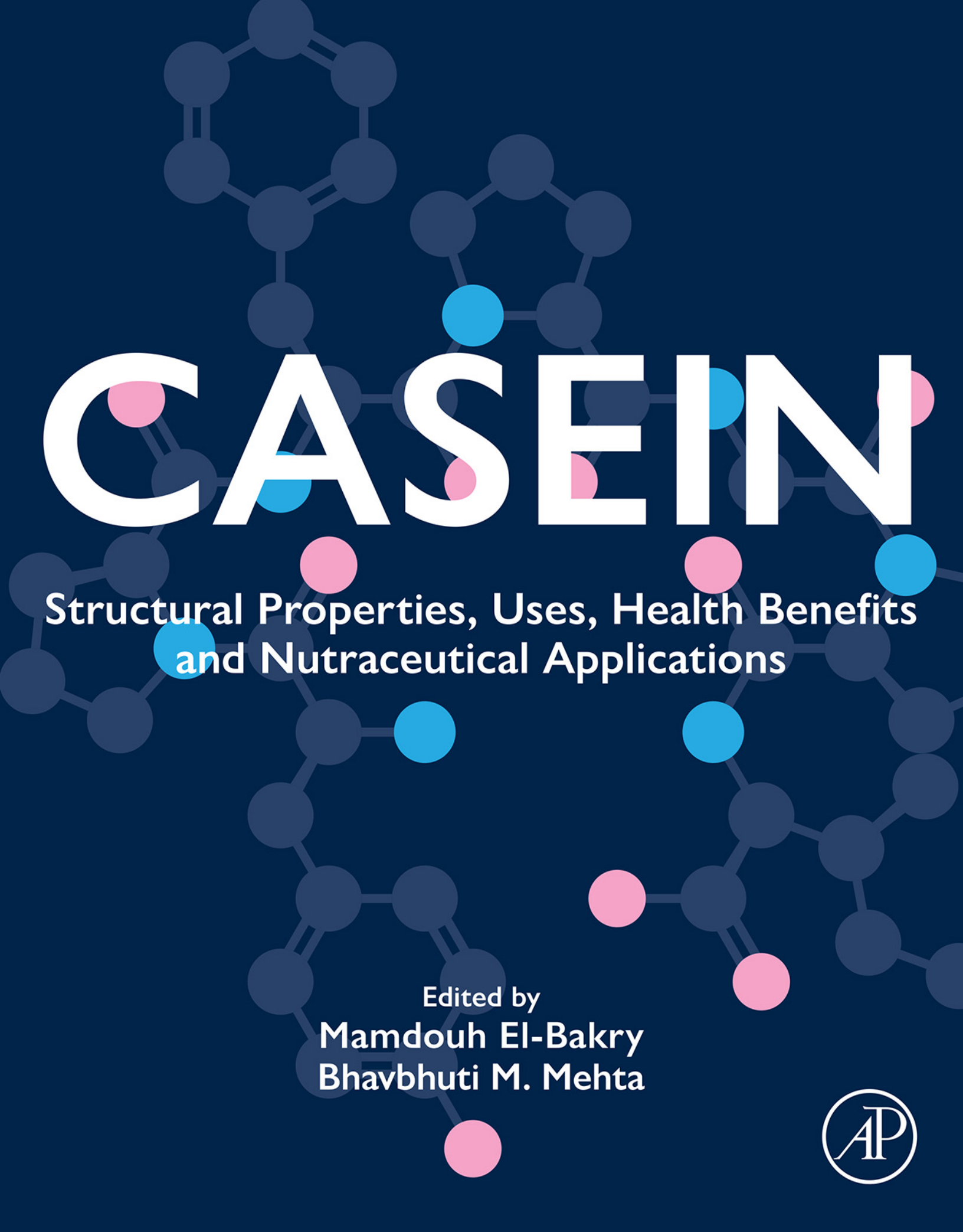


The background of the cover features a complex molecular structure graphic. It consists of numerous dark blue spheres connected by thin lines, forming a network of hexagonal and other geometric shapes. Interspersed within this network are several bright blue and pink spheres, which appear to be part of the molecular structure or represent specific atoms or functional groups. The overall effect is a scientific and modern aesthetic.

CASEIN

**Structural Properties, Uses, Health Benefits
and Nutraceutical Applications**

Edited by
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Bhavbhuti M. Mehta



Antimicrobial activities of casein

Javier Cifuentes¹, Paola Ruiz Puentes², Pablo Arbeláez²,
Carolina Muñoz-Camargo¹, Juan C. Cruz¹ and Luis H. Reyes³

¹Department of Biomedical Engineering, University of Los Andes, Bogota, Colombia ²Center for Research and Formation in Artificial Intelligence, University of Los Andes, Bogota, Colombia ³Department of Chemical and Food Engineering, University of Los Andes, Bogota, Colombia

19.1 Introduction

Dairy products have multiple benefits for boosting immunity and preventing infections (Darand et al., 2022). Specifically, milk reduces proinflammatory and oxidative stress factors (Al-Ayadhi et al., 2022; Darand et al., 2022), acts as an immunomodulator (Sibel Akalin, 2014), and lessens allergies, asthma, and respiratory infections (Perdijk et al., 2018). Among the proteins responsible for these benefits are caseins, whey proteins, lactoalbumins, and lactoglobulins (Ali et al., 2022; Darand et al., 2022). Dairy products have been recognized as an essential source of bioactive peptides. These are typically biologically occurring, short amino acid (AA) chains with covalent connections that are advantageous to human health (Ali et al., 2022). These peptides are produced in vivo by microbial fermentation or enzyme digestion of proteins and perform physiological functions in major bodily systems. Also, they can serve as signaling molecules and play crucial roles in biological processes and pathogenesis (Ali et al., 2022; Sibel Akalin, 2014). For dairy products, the following bioactivities have been attributed to peptides: antibacterial, antioxidant, angiotensin-converting enzyme inhibitory, cholesterol-lowering, antithrombotic, mineral bioavailability increase, cytomodulatory, and immunomodulatory (López-García et al., 2022; Sah et al., 2018; Sibel Akalin, 2014; Singh et al., 2023). In particular, milk-derived antimicrobial peptides (AMPs) are vital to the innate defense. They protect mucosal surfaces like the lungs and small intestine, which are continually exposed to various pathogens that might be risky to human health. Milk-derived AMPs are gaining popularity as a safe and effective alternative to antibiotics, with

the added benefit of shelf-life extension when incorporated into food products with different processing levels (Singh et al., 2023).

AMPs are common molecules in nature and have been identified in almost all forms of life comprehended, from single-celled bacteria to multicellular organisms. In different animals, including mammals, these peptides are critical components of the innate with a broad variety of antimicrobial activities (Bechinger, 2015). Individual AMPs' ability to engage with many targets, or multiple peptides' ability to interact with a single target, may potentially inhibit or delay the development of bacterial resistance. These peptides are therefore promising antibacterial agents with reduced rates of acquired resistance compared to conventional antibiotics (Bechinger and Gorr, 2017; Lázár et al., 2018). Their action depends on the interaction between the positively charged residues of the peptide and the negatively charged microbial membrane. Such an interaction is controlled by their ability to adopt an amphipathic conformation upon membrane binding. Nevertheless, their particular mechanisms of action depend on their structure, the propensity to aggregate, and interactions with the lipids of cellular membranes (Bechinger, 2015). These depend on the structure of the peptide, the peptide/lipid ratio, and the lipid membrane characteristics (Bechinger, 2015). In addition, a feature shared by AMPs is their multifunctionality, which produces numerous physiological effects when targeting distinct intracellular structures (Bechinger, 2015; Diener et al., 2016). In recent years, researchers have been increasingly interested in polypharmacological peptides (use of single drugs with multiple targets within a network) due to their demonstrated efficacy as

therapies for several illnesses (Cheng et al., 2019; Hao et al., 2017; Kula et al., 2020; Lázár et al., 2018; Li et al., 2020, 2017).

Casein comprises 80% of milk proteins and is the predominant protein fraction. Consequently, it is one of the principal sources of AMPs in dairy products (Khan et al., 2018; Singh et al., 2023). It has been demonstrated that casein hydrolysis generates various antimicrobial/bioactive peptides (Singh et al., 2023). As the primary source of AMPs in milk, this chapter describes several in vitro and in silico techniques to extract and validate the potential of the derived AMPs. Please add a sentence, explaining the next sections of the chapter. As a suggestion: “Two main production methods will be discussed in detail; the empirical and computational approach.”

19.2 Antimicrobial peptides: production methods

Both empirical and computational approaches have been utilized extensively in the discovery of AMPs (Capriotti et al., 2016). The empirical method is based on the in vitro hydrolysis of casein sources by proteolytic enzymes or microbial fermentation methods (Mohanty et al., 2016). In contrast, the computational approach employs several bioinformatic and computational approaches to uncover and predict the biological function of peptide candidates in silico (Capriotti et al., 2016; Liu et al., 2015).

19.2.1 Empirical approach

The first and most conventional method, the empirical approach, consists of a series of steps in which a casein precursor source is extracted from milk or other dairy products and then is treated enzymatically to obtain small fragments. These fragments can be isolated, characterized, and assessed to identify their biological role (Capriotti et al., 2016). In this process, the peptides might be generated utilizing three distinct methods. The first approach involves enzymatic hydrolysis with digestive enzymes such as pepsin, trypsin, and chymotrypsin (Mohanty et al., 2016). In contrast, the second technique utilizes proteolytic enzymes derived from nongastrointestinal sources such as animals, plants, and microorganisms (El-Sayed and Awad, 2019). It is possible to highlight plasmin, chymosin, and alcalase among these enzymes (Mohanty et al., 2016). Finally, the third method, termed the fermentation approach, consists of the cleavage of casein sources through the activity of microbial strains-produced proteolytic enzymes (Capriotti et al., 2016; Mohanty et al., 2016). Lactic acid

bacteria (LAB) are the microbial strains used the most to obtain bioactive peptides. Examples of such LAB strains include *Lactococcus lactis*, *Lactobacillus helveticus*, and *Lactobacillus acidophilus* (El-Sayed and Awad, 2019; Mohanty et al., 2016). Fig. 19.1 shows the schematic representation of the empirical approach with the different peptide-release methods.

Schematic representation of the different methods based on the empirical approach to obtain casein-derived AMPs (created with BioRender.com). Milk is obtained from different animal origins, and then, casein fraction is isolated to obtain AMPs candidates (casein proteins are shown: α_{S1} -casein, α_{S2} -casein, β -casein, and κ -casein). The methods for AMPs obtention are presented: enzymatic hydrolysis-based strategies (*Hydrolysis*: nongastrointestinal-derived proteolytic enzymes and *Digestion*: gastrointestinal-derived proteolytic enzymes) and in vitro fermentation by using microorganisms.

19.2.1.1 Enzymatic hydrolysis

19.2.1.1.1 Gastrointestinal-derived proteolytic enzymes

In vitro enzymatic hydrolysis by digestive enzymes models in vivo gastrointestinal protein digestion (Park and Nam, 2015). In this approach, peptides are generated by the activity of digestive enzymes found naturally in the digestive tracts of mammals (Mohanty et al., 2016). This technique has contributed to the discovery of several peptides with exceptionally high antibacterial activity. Particularly pepsin, the most important digesting enzyme, has enabled the identification of 34 AMPs with unique action against gram-negative and gram-positive bacteria (Singh et al., 2023). For example, Recio and Visser (1999) reported for the first time the identification of two distinct bovine α_{S2} -casein-derived AMPs, LKKISQRYQKFALPQY, and VYQHQAAMKPWIQPKTKVIPVRYL. These peptides show potent bactericidal activity against *Escherichia coli* and numerous gram-positive strains, including *Listeria innocua*, *Bacillus cereus*, *Mariniluteicoccus flavus*, and *Streptococcus thermophilus*. Similarly, McCann and colleagues reported for the first time the identification of the bovine α_{S1} -casein peptide, LRLKKYKVPQL (Cp1), a potent AMP with important activity against *Bacillus subtilis*, *L. innocua*, *Listeria monocytogenes*, *Citrobacter freundii*, *E. coli*, *Salmonella enteritidis*, and *Salmonella typhimurium* (McCann et al., 2006). Cp1 has been intensively investigated over the past few years, revealing extraordinary activity in an increasing number of bacterial strains, including highly pathogenic strains such as *Salmonella pullorum*, *Salmonella enterica*, *Staphylococcus aureus*, and *L. monocytogenes* (Hou et al., 2018; Tang et al., 2015). In addition, Hou et al. (2018) demonstrated that Cp1 possessed broad-spectrum antimicrobial activity and high

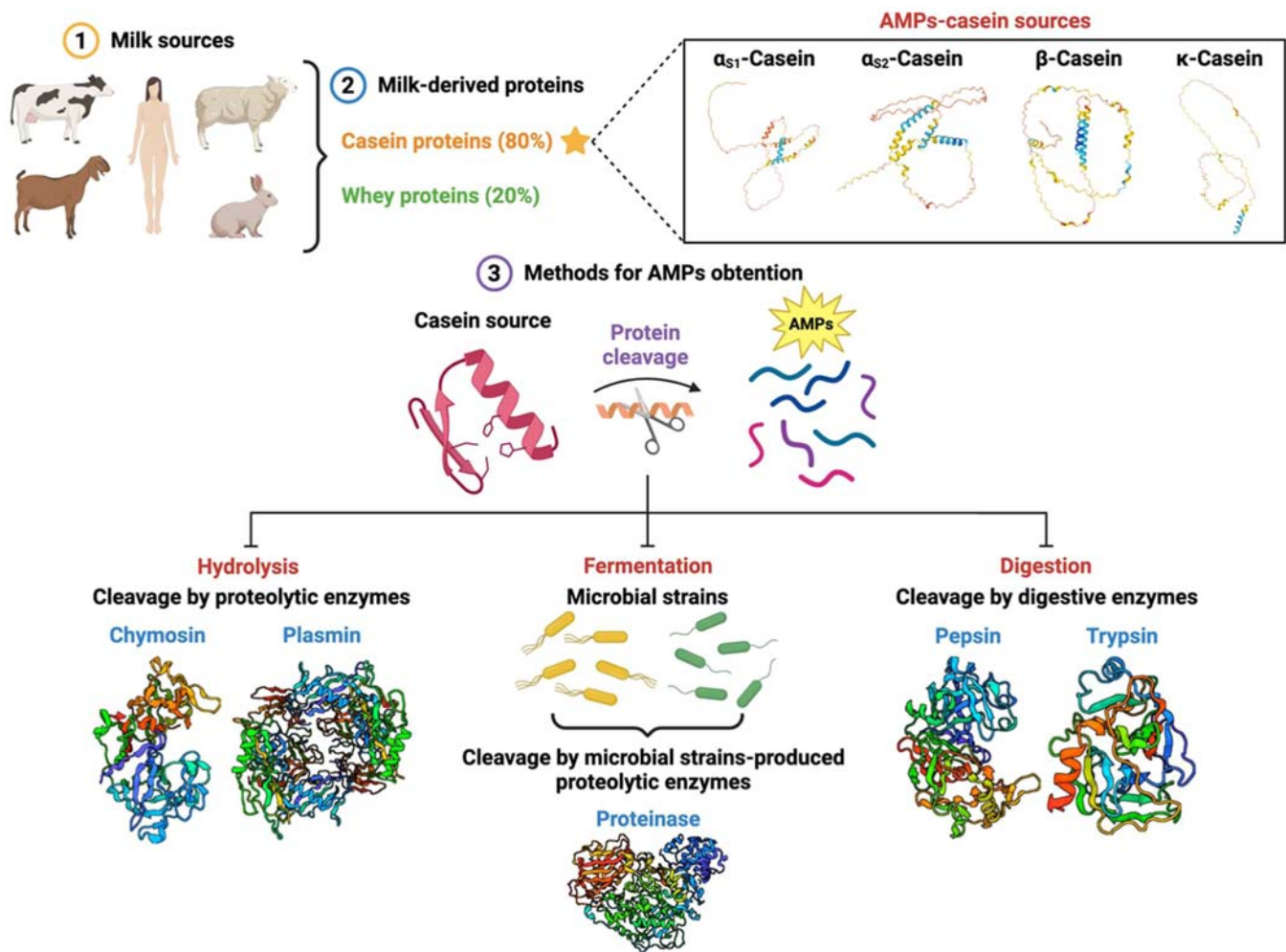


FIGURE 19.1 Schematic representation of the different methods based on the empirical approach to obtain casein-derived AMPs.

biocompatibility, as indicated by its low hemolytic activity and low cytotoxicity toward human cells. These findings revealed Cp1 potential as a versatile replacement for conventional food preservatives. On the other hand, Wan et al. (2021) confirmed the antimicrobial activity of pepsin-mediated bovine-casein hydrolysates against *Helicobacter pylori*. The authors hypothesized that the peptides casecidin 17 (β -casein-derived peptide) and β -casein 207–224 are responsible for the anti-*H. pylori* properties. However, individual testing is yet to be conducted to confirm the hypothesis (Wan et al., 2021). Casecidin 17 was found as a naturally occurring peptide in bovine colostrum, making it a promising option for treating *H. pylori*-associated disorders such as gastritis, peptic ulcer, and even stomach cancer (Birkemo et al., 2009; Wan et al., 2021). In addition, goat β -casein-derived peptides, such as TEDELQDKIHFP, REQEELNV, and PFTGPIPNLPO, have demonstrated critical antibacterial activity against *E. coli* and a pathogenic strain *B. cereus*. This strain has shown resistance to cooking and

pasteurization, the two most popular food industry sterilization methods (Almaas et al., 2011; Esmaeilpour et al., 2016). Table 19.1 presents all the casein-derived peptides obtained by pepsin hydrolysis.

Trypsin, the second-most-used digestive enzyme, has also proven a significant ability to generate casein-derived AMPs (Singh et al., 2023). It has been reported that around four casein-derived peptides identified through trypsin hydrolysis possess antibacterial action against various strains. For instance, McClean et al. (2014) demonstrated the action of the bovine α_{S1} -casein peptide TTMLPW not only against gram-positive and gram-negative bacteria but also against the pathogenic yeast *Candida albicans* (McClean et al., 2014). In addition, Baranyi et al. (2003) presented the discovery of the following AMPs isolated from rabbit milk (Baranyi et al., 2003), the α_{S2} -casein-derived peptide FHLGHLK and the β -casein-derived peptides HVEQLLR and ILPFIQSLFPFAER. All peptides presented activity against *B. subtilis*, and ILPFIQSLFPFAER also exhibited

TABLE 19.1 Casein-derived antimicrobial peptides (AMPs), sequences, release methods, and target strains.

Casein source	AMP sequence	Fragment position	AA number	Species origin	Released by	Antimicrobial activity	References
αs1-Casein	YLEQLLR	109–115	7	<i>Bos taurus</i>	Virtual screening	<i>Bacillus subtilis</i> , <i>Escherichia coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	VLNENLLR	30–37	8	<i>B. taurus</i>	Fermentation— <i>Lactobacillus acidophilus</i> DPC6026/ Fermentation— <i>Bacillus cereus</i> and <i>Bacillus thuringiensis</i>	<i>E. coli</i> , <i>Cronobacter sakazakii</i> , <i>Listeria innocua</i> , <i>Lactobacillus bulgaricus</i> , and <i>Streptococcus mutans</i> / <i>C. sakazakii</i> / <i>C. sakazakii</i> and <i>Cronobacter muytjensii</i>	Hayes et al. (2006) , Kent et al. (2012) , and Norberg et al. (2012)
αs1-Casein	TTMPLW	209–214	6	<i>B. taurus</i>	Direct synthesis (discovered by using trypsin)	<i>E. coli</i> , <i>Staphylococcus aureus</i> , <i>Micrococcus luteus</i> , and <i>Candida albicans</i>	McClean et al. (2014)
αs1-Casein	SDIPNPIGSENSEK	195–208	14	<i>B. taurus</i>	Fermentation— <i>L. acidophilus</i> DPC6026	<i>L. innocua</i>	Hayes et al. (2006)
αs1-Casein	RPKHPIKHQGLPQEVNENLLR FFVAPFPEVFGKEK	16–52	37	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	RPKHPIKHQGLPQEVNEN LLRFFVAPFPEVFGKEK	16–51	36	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	RPKHPIKHQGLPQEVNEN NLLRFFVAPFPEVFGK	16–49	34	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	RPKHPIKHQGLPQE VLNENLLRFF	16–39	24	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	RPKHPIKHQGLPQE VLNENLLRF	16–38	23	<i>B. taurus</i>	Chymosin/not applicable—fresh bovine colostrum	<i>S. aureus</i> , <i>Streptococcus pyogenes</i> , and <i>Listeria monocytogenes</i> / <i>E. coli</i> and <i>C. sakazakii</i>	Lahov and Regelson (1996) and Birkemo et al. (2009)
αs1-Casein	RPKHPIK	16–22	7	<i>B. taurus</i>	Not applicable—identified from bovine milk—caciocavallo cheese	<i>Latilactobacillus sakei</i> A15, <i>E. coli</i> K12, and <i>Bacillus megaterium</i> F6	Rizzello et al. (2005)
αs1-Casein	LRLKKYKVPQL	114–124	11	<i>B. taurus</i>	Pepsin/pepsin/direct synthesis	<i>B. subtilis</i> , <i>L. innocua</i> , <i>L. monocytogenes</i> , <i>Citrobacter freundii</i> , <i>E. coli</i> , <i>S. enteritidis</i> , and <i>Salmonella typhimurium</i> / <i>B. subtilis</i> , <i>S. aureus</i> , <i>S. pneumoniae</i> , <i>E. coli</i> , <i>S. dysenteriae</i> , <i>S. typhimurium</i> , and <i>Salmonella enterica</i> / <i>E. coli</i> , <i>Salmonella pullorum</i> , <i>S. enterica</i> , <i>S. aureus</i> , and <i>L. monocytogenes</i>	McCann et al. (2006) , Tang et al. (2015) , and Hou et al. (2018)
αs1-Casein	LGYLEQLLRL	107–116	10	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	LEQLLRLKKY	110–119	10	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)

αs1-Casein	IKHQGLPQEV	21–30	10	<i>B. taurus</i>	Mixture of proteases	<i>E. coli</i> and <i>B. subtilis</i>	Elbarbary et al. (2012)
αs1-Casein	IKHQGLPQE	21–29	9	<i>B. taurus</i>	Fermentation— <i>L. acidophilus</i> DPC6026/fermentation— <i>B. cereus</i> and <i>B. thuringiensis</i>	<i>E. coli</i> , <i>C. sakazakii</i> , <i>L. innocua</i> , <i>L. bulgaricus</i> , and <i>S. mutans</i> / <i>C. sakazakii</i> / <i>C. sakazakii</i> and <i>C. muytjensii</i>	Hayes et al. (2006) , Kent et al. (2012) , and Norberg et al. (2012)
αs1-Casein	HIQKEDVPSERYLGYL EQLRLKKYK	95–120	26	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	HIQKEDVPSERYLGYL EQLRLKK	95–118	24	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	HIQKEDVPSERYLG YLEQLRLK	95–117	23	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs1-Casein	FHLGHLK	19–25	7	<i>Oryctolagus cuniculus</i>	Trypsin	<i>B. subtilis</i>	Baranyi et al. (2003)
αs2-Casein	WIQPKTKVIPYVRYL	208–222	15	<i>B. taurus</i>	Direct synthesis (discovered by using pepsin)	<i>C. sakazakii</i> and <i>L. monocytogenes</i>	Alvarez-Ordóñez et al. (2013)
αs2-Casein	VYQHQAAMKPWIQ PKTKVIPYVRYL	198–222	25	<i>B. taurus</i>	Pepsin/pepsin/direct synthesis/direct synthesis/direct synthesis	<i>B. subtilis</i> , <i>L. innocua</i> , <i>L. monocytogenes</i> , <i>C. freundii</i> , <i>E. coli</i> , <i>S. enteritidis</i> , and <i>S. typhimurium</i> / <i>E. coli</i> , <i>L. innocua</i> , <i>B. cereus</i> , <i>M. flavus</i> , and <i>S. thermophilus</i> / <i>E. coli</i> and <i>Staphylococcus carnosus</i> / <i>E. coli</i> and <i>S. carnosus</i> / <i>C. sakazakii</i> and <i>L. monocytogenes</i>	McCann et al. (2006) , Recio and Visser (1999) , López-Expósito et al. (2008a, 2008b) , Alvarez-Ordóñez et al. (2013)
αs2-Casein	TVYQHQAAMKPWI QPKTKVIPYVRYL	197–222	26	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs2-Casein	TKVIPYVRYL	213–222	10	<i>B. taurus</i>	Direct synthesis (discovered by using pepsin)	<i>C. sakazakii</i> and <i>L. monocytogenes</i>	Alvarez-Ordóñez et al. (2013)
αs2-Casein	TKLTEEEKNRLNFK KISQRYQKFALPQYLK	166–196	31	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
αs2-Casein	TKKTKLTEEEKNRL	163–176	14	<i>B. taurus</i>	Direct synthesis (discovered by using chymotrypsin)	<i>B. cereus</i> , <i>S. aureus</i> , and <i>L. monocytogenes</i>	Sistla (2013)
αs2-Casein	QKFALPQYLKTVYQHQA AMKPWIQPKTKVIPYVRYL	187–222	36	<i>B. taurus</i>	Chymosin	<i>L. innocua</i> and <i>B. subtilis</i>	McCann et al. (2005)
αs2-Casein	QKALNEINQF	94–103	10	<i>B. taurus</i>	Direct synthesis (discovered by using chymotrypsin)	<i>B. cereus</i> , <i>S. aureus</i> , <i>L. monocytogenes</i> , and <i>Helicobacter pylori</i>	Sistla (2013)
αs2-Casein	LKTVYQHQAAMKPWI QPKTKVIPYVRYL	195–222	28	<i>B. taurus</i>	Chymosin	<i>B. cereus</i> , <i>B. subtilis</i> , and <i>L. innocua</i>	McCann et al. (2005)

(Continued)

TABLE 19.1 (Continued)

Casein source	AMP sequence	Fragment position	AA number	Species origin	Released by	Antimicrobial activity	References
α S2-Casein	LKKISQRYQKFALPQYLKT VYQHQAAMKWPQPKTKVIPYVRYL	179–222	44	<i>B. taurus</i>	Chymosin	<i>B. subtilis</i> , <i>L. innocua</i> , and <i>L. monocytogenes</i> 2310	McCann et al. (2005)
α S2-Casein	LKKISQRYQKFALPQY	179–194	16	<i>B. taurus</i>	Pepsin	<i>E. coli</i> , <i>L. innocua</i> , <i>B. cereus</i> , <i>M. flavus</i> and <i>S. thermophilus</i>	(Recio and Visser, 1999)
α S2-Casein	KTVYQHQAAMKWP IQPKTKVIPYVRYL	196–222	27	<i>B. taurus</i>	Chymosin	<i>B. subtilis</i> , <i>L. innocua</i> , <i>L. monocytogenes</i> 2310, and <i>S. typhimurium</i>	McCann et al. (2005)
α S2-Casein	KTKLTEEEKNRLNFLKKIS QRYQKFALPQYLKTVYQHQA	165–203	39	<i>B. taurus</i>	Not applicable— identified from bovine milk	<i>E. coli</i> and <i>S. carnosus</i>	Zucht et al. (1995)
α S2-Casein	KKISQRYQKFALPQYLKTVYQHQA	180–203	24	<i>B. taurus</i>	Mixture of proteases	<i>E. coli</i> and <i>B. subtilis</i>	Elbarbary et al. (2012)
α S2-Casein	KAMKWPQPKTKVIPYVRYL	203–222	20	<i>B. taurus</i>	Direct synthesis (discovered by using pepsin)	<i>C. sakazakii</i> and <i>L. monocytogenes</i>	Alvarez-Ordóñez et al. (2013)
α S2-Casein	KAMKWPQPKTKVIP	203–217	15	<i>B. taurus</i>	Direct synthesis (discovered by using pepsin)	<i>C. sakazakii</i> and <i>L. monocytogenes</i>	Alvarez-Ordóñez et al. (2013)
α S2-Casein	KAMKWP	203–208	6	<i>B. taurus</i>	Mixture of proteases	<i>E. coli</i> and <i>B. subtilis</i>	Elbarbary et al. (2012)
α S2-Casein	IVLNPWDQVK	119–128	10	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
α S2-Casein	IQPKTKVIPYVR	209–220	12	<i>B. taurus</i>	Mixture of proteases	<i>E. coli</i> and <i>B. subtilis</i>	Elbarbary et al. (2012)
α S2-Casein	ALPQYLKTVYQHQAAMK PWIQPKTKVIPYVRYL	190–222	33	<i>B. taurus</i>	Chymosin	<i>B. subtilis</i> , <i>L. innocua</i> , <i>L. monocytogenes</i> 2310, and <i>S. typhimurium</i>	McCann et al. (2005)
α S2-Casein	SSSEESII	23–30	8	<i>B. taurus</i>	Serine metalloprotease produced by <i>L. lactis</i> subsp <i>lactis</i> BR16	<i>L. innocua</i> , <i>M. luteus</i> , <i>E. coli</i> , and <i>S. enteritidis</i>	Bougherra et al. (2017)
α S2-Casein	KTVDMESTEVFTKKTK LTEEEKNRLNFLKK	152–181	30	<i>B. taurus</i>	Direct synthesis (discovered by virtual screening)	<i>B. subtilis</i>	Liu et al. (2020)
α S2-Casein	VDQHQAAMKPWT QPKTNAIPYVRYL	199–223	25	<i>Ovis aries</i>	Pepsin	<i>S. carnosus</i> and <i>L. innocua</i>	López-Expósito et al. (2006a)
α S2-Casein	LKKISQ	180–185	6	<i>O. aries</i>	Pepsin	<i>S. carnosus</i> , <i>S. epidermidis</i> , and <i>L. innocua</i>	López-Expósito et al. (2006a)
α S2-Casein	LKKISQYYQKFAWPQYL	180–196	17	<i>O. aries</i>	Pepsin	<i>E. coli</i> , <i>S. carnosus</i> , <i>S. epidermidis</i> , <i>E. faecalis</i> , and <i>L. innocua</i>	López-Expósito et al. (2006a)
α S2-Casein	PYVRYL	217–222	6	<i>O. aries</i>	Pepsin	<i>S. carnosus</i> , <i>S. epidermidis</i> , and <i>L. innocua</i>	López-Expósito et al. (2006a)

β -Casein	YQEPVLGPVRGPFPIIV	208–224	17	<i>B. taurus</i>	Not applicable—fresh bovine colostrum	<i>E. coli</i>	Birkemo et al. (2009)
β -Casein	YPVEPF	129–134	6	<i>B. taurus</i>	Fermentation— <i>B. cereus</i> RC6 strain	<i>B. cereus</i> ATCC 11778, <i>L. monocytogenes</i> DISTAM MACa1, and <i>B. thuringiensis</i> USDA HD22	Ouertani et al. (2018)
β -Casein	VPYPQRDMPIQAFI	193–206	14	<i>B. taurus</i>	Virtual screening	<i>B. subtilis</i> , <i>E. coli</i> NEB 5 alpha, and <i>E. coli</i> ATCC 25922	Liu et al. (2015)
β -Casein	VLPVPQKAVPYPQR	185–198	14	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	VLPVPQK	185–191	7	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	VKEAMAPK	113–120	8	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	RINKK	40–44	5	<i>B. taurus</i>	Pepsin	<i>E. coli</i>	López-Expósito et al. (2006b)
β -Casein	HKEMPFPK	121–128	8	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	EMPFPK	123–128	6	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	EAMAPKHK	115–122	8	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	EAMAPK	115–120	6	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2016)
β -Casein	AVPYPQR	192–198	7	<i>B. taurus</i>	Mixture of proteases/plasmin	<i>E. coli</i> and <i>B. subtilis</i> / <i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Elbarbary et al. (2012) and Sedaghati et al. (2016)
β -Casein	DVENLHLPLPL	144–154	11	<i>B. taurus</i>	Fermentation— <i>L. helveticus</i> LH-2 and <i>L. acidophilus</i> La-5	<i>S. typhimurium</i>	Ali and LaPointe (2022)
β -Casein	GVSKVKEAMAPKH	109–121	13	<i>B. taurus</i>	Fermentation— <i>L. helveticus</i> LH-2 and <i>L. acidophilus</i> La-5	<i>S. typhimurium</i>	Ali and LaPointe (2022)
β -Casein	SSSEESITRIN	32–42	11	<i>B. taurus</i>	Fermentation— <i>L. helveticus</i> LH-2 and <i>L. acidophilus</i> La-5	<i>S. typhimurium</i>	Ali and LaPointe (2022)
β -Casein	ELNVPGEIVES	20–30	10	<i>B. taurus</i>	Fermentation— <i>L. helveticus</i> LH-2 and <i>L. acidophilus</i> La-5	<i>S. typhimurium</i>	Ali and LaPointe (2022)
β -Casein	YQEPVLGPVRGPFPI	206–220	15	<i>Capra hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i>	Almaas et al. (2011) and Birkemo et al. (2009)
β -Casein	TEDELQDKIHPF	56–67	12	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> and <i>B. cereus</i>	Almaas et al. (2011)

(Continued)

TABLE 19.1 (Continued)

Casein source	AMP sequence	Fragment position	AA number	Species origin	Released by	Antimicrobial activity	References
β -Casein	REQEELNV	16–23	8	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> and <i>B. cereus</i>	Almaas et al. (2011)
β -Casein	PVVVPFLQPE	96–106	11	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i>	Almaas et al. (2011)
β -Casein	PFTGPIPNLQP	76–87	12	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> and <i>B. cereus</i>	Almaas et al. (2011)
β -Casein	MHQPPQPL	159–166	8	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i>	Almaas et al. (2011)
β -Casein	KETMVPK	114–120	7	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i>	Almaas et al. (2011)
β -Casein	QELLLNPHTQ IYPVTQPLAPVHNPISV	200–226	27	<i>Homo sapiens</i>	Proteinase produced by <i>L. helveticus</i> PR4	<i>L. bulgaricus</i> , <i>L. casei</i> , <i>L. sanfranciscensis</i> , <i>E. faecium</i> , <i>L. lactis</i> , <i>B. megaterium</i> , <i>E. coli</i> , <i>L. innocua</i> , and <i>S. aureus</i>	Minervini et al. (2003)
β -Casein	LLNQELLLNPHTQIYPV	197–213	17	<i>H. sapiens</i>	Not applicable—identified from human milk	<i>E. coli</i> , <i>S. aureus</i> , and <i>Yersinia enterocolitica</i>	Fu et al. (2017)
β -Casein	YPVTQPLAPVHNPIIS	211–225	15	<i>H. sapiens</i>	Not applicable—identified from human milk	<i>E. coli</i> and <i>Y. enterocolitica</i>	Wang et al. (2020)
β -Casein	ILPFIQSLFPFAER	64–77	14	<i>O. cuniculus</i>	Trypsin	<i>E. coli</i> and <i>B. subtilis</i>	Baranyi et al. (2003)
β -Casein	HVEQLLR	50–56	7	<i>O. cuniculus</i>	Trypsin	<i>B. subtilis</i>	Baranyi et al. (2003)
κ -Casein	YYQQKPVA	63–70	8	<i>B. taurus</i>	Pepsin	<i>E. coli</i> and <i>S. carnosus</i>	López-Expósito et al. (2006b)
κ -Casein	YVL	51–53	3	<i>B. taurus</i>	Pepsin	<i>S. marcescens</i> , <i>L. innocua</i> , and <i>S. carnosus</i>	López-Expósito et al. (2006b)
κ -Casein	VQVTSTAV	183–190	8	<i>B. taurus</i>	Pepsin	<i>L. innocua</i> and <i>S. carnosus</i>	López-Expósito et al. (2006b)
κ -Casein	VESTVATL	160–167	8	<i>B. taurus</i>	Pepsin	<i>E. coli</i> , <i>L. innocua</i> , and <i>S. carnosus</i>	López-Expósito et al. (2006b)
κ -Casein	TEAVESTVATL	157–167	11	<i>B. taurus</i>	Mixture of proteases	<i>E. coli</i> and <i>B. subtilis</i>	Elbarbary et al. (2012)
κ -Casein	STVATL	162–167	6	<i>B. taurus</i>	Pepsin	<i>E. coli</i> , <i>L. innocua</i> , and <i>S. carnosus</i>	López-Expósito et al. (2006b)

κ-Casein	PAAVRSPAQLQ	85–96	12	<i>B. taurus</i>	Pepsin	<i>E. coli</i>	López-Expósito et al. (2006b)
κ-Casein	MMK	1–3	3	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2014)
κ-Casein	MAIPPKKNQDKTEIPTI NTIASGEPTSTPTTEAVE STVATLEDSPEVIESPPEI NTVQVTSTAV	127–190	64	<i>B. taurus</i>	Chymosin	<i>E. faecalis</i> and <i>S. mutans</i>	Liu et al. (2012) and Malkoski et al. (2001)
κ-Casein	MAIPPKKNQDKTEIPTINT	127–145	19	<i>B. taurus</i>	Chymosin	<i>E. coli</i>	Robitaille et al. (2012)
κ-Casein	IQY	49–51	3	<i>B. taurus</i>	Pepsin	<i>E. coli</i>	López-Expósito et al. (2006b)
κ-Casein	IAK	43–45	3	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2014)
κ-Casein	FSDKIAK	39–45	7	<i>B. taurus</i>	Pepsin	<i>E. coli</i> , <i>L. innocua</i> , and <i>S. carnosus</i>	López-Expósito et al. (2006b)
κ-Casein	FFSDK	38–42	5	<i>B. taurus</i>	Plasmin	<i>E. coli</i> , <i>S. aureus</i> , <i>L. casei</i> , and <i>L. acidophilus</i>	Sedaghati et al. (2014)
κ-Casein	EIPT	139–142	4	<i>B. taurus</i>	Pepsin	<i>E. coli</i> and <i>S. marcescens</i>	López-Expósito et al. (2006b)
κ-Casein	AVESTVATLEDSPEVIESPPE	159–179	21	<i>B. taurus</i>	Chymosin	<i>S. mutans</i>	Malkoski et al. (2001)
κ-Casein	MAIPPKKDQDKTEVPAINT	127–145	19	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> , <i>B. cereus</i> , and <i>L. monocytogenes</i>	Almaas et al. (2011)
κ-Casein	IVNTVDNPEASSE	162–174	13	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> and <i>B. cereus</i>	Almaas et al. (2011)
κ-Casein	PTVHSTPTTE	151–160	10	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> , <i>B. cereus</i> , and <i>L. monocytogenes</i>	Almaas et al. (2011)
κ-Casein	PPKKDQDKTEVPA	130–142	13	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> , <i>B. cereus</i> , and <i>L. monocytogenes</i>	Almaas et al. (2011)
κ-Casein	ASAEPTVH	147–154	8	<i>C. hircus</i>	Pepsin and human gastrointestinal enzymes/juices	<i>E. coli</i> , <i>B. cereus</i> , and <i>L. monocytogenes</i>	Almaas et al. (2011)
κ-Casein	YQRRPAIAINNPYVPRTYY ANPAVVVRPHAQIPQRQYLPN SHPPTVVRPNLHPSF	63–117	55	<i>H. sapiens</i>	Pepsin	<i>E. coli</i>	Liepke et al. (2001)

Modified From Nielsen, S.D., Beverly, R.L., Qu, Y., Dallas, D.C., 2017. Milk bioactive peptide database: a comprehensive database of milk protein-derived bioactive peptides and novel visualization. *Food Chemistry* 232, 673–682. <https://doi.org/10.1016/j.foodchem.2017.04.056>.

activity against *E. coli*. Ultimately, chymotrypsin was used to identify two distinct casein-derived peptides with outstanding antibacterial activity. These peptides were reported by Sistla (2013) and were derived from bovine α_{S2} -casein-derived AMPs, QKALNEINQF, and TKKTKLTEEEKNRL (Sistla, 2013). The peptides showed significant antibacterial activity against the pathogenic strains *B. cereus*, *S. aureus*, and *L. monocytogenes*. In addition, QKALNEINQF exhibits an important activity against *H. pylori*. Sistla (2013) presented all the casein-derived peptides obtained upon hydrolysis with gastrointestinal-derived proteolytic enzymes.

19.2.1.1.2 Nongastrointestinal-derived proteolytic enzymes

Several works have reported the discovery of casein-derived peptides through the activity of enzymes of various origins, in addition to the most common proteolytic enzymes produced from the gastrointestinal tract. Mudgil et al. evaluated cow and camel casein hydrolysates, produced from alcalase (*Bacillus licheniformis*) and protease (*Streptomyces griseus*), against various *Candida* species (Mudgil et al., 2022). Both casein hydrolysates showed high antifungal potency against the pathogenic strains *C. albicans*, *Candida krusei*, and *Candida parapsilosis*. Furthermore, camel milk-derived hydrolysate showed more potent antifungal activity than the evaluated cow milk-derived hydrolysate (Mudgil et al., 2022). However, this study failed to identify nonindividual peptides. Consequently, future research must focus on identifying the antimicrobial molecules linked with the observed antifungal action. Moreover, Zhao et al. reported on the discovery of a new AMP produced from the casein fraction of buffalo milk. This study produced a casein hydrolysate using a protease isolated from the *Dregea sinensis* Hemsl plant (Zhao et al., 2020). The peptide BCp12 (YLGYLEQLRLK) showed relevant activity against the gram-positive bacteria *S. aureus* and *L. monocytogenes* and the gram-negative bacteria *E. coli* and *S. typhimurium*. In addition, the authors demonstrated the high biocompatibility of the peptide through low hemolytic activity and remarkable cytocompatibility against human epidermal keratinocytes (HaCat cells). This peptide is an up-and-coming candidate for developing innovative treatments against *S. aureus*-infected skin diseases (Zhao et al., 2020). Similarly, recent works by Shi et al. (2022) and Li et al. (2022b) explored the *S. aureus*-inhibition mechanism as well as the antibiofilm properties of the BCp12 peptide. In all investigations, the high antibacterial activity of BCp12 was validated, revealing this peptide as a candidate for novel antimicrobial developments in the pharmaceutical and food industries (Li et al., 2022b; Shi et al., 2022). The sequence of

BCp12 corresponds to the residues 106–117 of bovine α_{S1} -casein. Also, this sequence is included in three different AMPs previously discovered by Liu et al. using the virtual screening approach (Liu et al., 2015) (see Table 19.1). On the other hand, around 20 peptides were discovered by employing plasmin (11) and chymosin (9) as proteolytic enzymes. These peptides include Isracidin and Kappacin, two of the most widely studied casein-derived AMPs. Isracidin, RPKHPIKHQGLPQEVLENLLRF, is a bovine α_{S1} -casein-derived peptide with remarkable antibacterial activity against several gram-negative and gram-positive bacteria, including *S. aureus*, *Streptococcus pyogenes*, *L. monocytogenes*, *E. coli*, and *Cronobacter sakazakii* (Birkemo et al., 2009; Lahov and Regelson, 1996). Isracidin sequence has been identified in the structure of four different casein-derived AMPs (see Table 19.1; Liu et al., 2015). This peptide was discovered to be naturally present in bovine colostrum. Its immunomodulatory properties make it a particularly intriguing nutraceutical agent candidate (Girija, 2018). Finally, Kappacin (κ -casein 127–190) is the non-glycosylated, phosphorylated form of the highly investigated bovine casein-macro peptide. Kappacin is the largest casein-derived AMP (64 residues) that shows significant antibacterial activity against several pathogenic strains (Dashper et al., 2007). Notably, its applicability as a potential oral care product ingredient has been intensively researched. This is due to its capacity to suppress the growth of oral plaque-forming and cariogenic bacteria in a biofilm, including *S. mutans*, *Porphyromonas gingivalis*, and *Actinomyces naeslundii*. Dashper et al. (2007) presented all the casein-derived peptides obtained upon hydrolysis with nongastrointestinal-derived proteolytic enzymes.

19.2.1.2 In vitro fermentation

More than 40% of the peptide linkages in the structure of α_{S1} - and β -casein can be hydrolyzed by proteolytic enzymes generated by LAB, yielding numerous peptides with bioactive activities (El-Sayed and Awad, 2019). On this basis, LAB strains have been widely used to produce bioactive peptides from casein fractions or directly from dairy products obtained through a fermentation process. This method is also used in the food sector to produce cheese, yogurt, and kashk, among other fermented dairy products (Folliero et al., 2022; Novak et al., 2021). The choice of particular LAB strains directly defines the bioactive peptide content of dairy products. Fermentation with various LAB strains, such as *L. acidophilus*, *Bacillus thuringiensis*, and *L. helveticus*, has led to the identification of approximately 10 casein-derived bioactive peptides with independently confirmed antibacterial activity. For instance, Nebbia et al. reported using *L. lactis*, *L. rhamnosus*, and *L. helveticus* as

fermentation agents to generate casein-derived bioactive peptides. This work obtained several previously reported AMPs, including TKTKLTETEEKNRL and QKALNEINQF, two bovine α_{S2} -casein-derived peptides with potent activity against *B. cereus*, *S. aureus*, and *L. monocytogenes* (Nebbia et al., 2020). These peptides were first decrypted by using chymotrypsin (Sistla, 2013). However, this work demonstrated the successful use of *L. rhamnosus* and *L. helveticus* to release both peptides. Similarly, authors reported the peptides LEQLRLKKY obtained by *L. lactis* and *L. helveticus* from α_{S1} -casein and YQEPVLGPVRGPFPI (the well-known caseicin 15) by *L. rhamnosus* from β -casein (Nebbia et al., 2020). Muhialdin and Alboory also presented the Fermentation of Iraqi camel milk from *L. plantarum* (Muhialdin and Alboory, 2018). The fraction with the most potent antimicrobial activity against *Enterococcus faecalis*, *Shigella dysenteriae*, *S. aureus*, and *E. coli* yielded several peptides derived from β -, α_{S1} - and α_{S2} -casein (Muhialdin and Alboory, 2018). However, nonindividual peptides were examined; therefore, additional research is required to identify and test the associated antimicrobial molecules individually.

On the other hand, Novak et al. conducted a fascinating study in which they established a novel strategy for enhancing the bioactive peptide content of dairy products by utilizing LAB consortiums (Novak et al., 2021). In this study, the typical use of one or two LAB strains for cheese manufacturing was replaced with a consortium of LAB strains. This consortium consisted of three different Lactobacillus strains: *L. brevis*, *L. fermentum*, and *L. plantarum*, and one *Enterococcus* strain, *E. faecium*. The authors revealed that the LAB consortium dramatically improves the number and quantity of casein-derived peptides available. Specifically, this strategy promoted the production of bioactive peptides produced from β -casein, including some exhibiting antibacterial activity (Novak et al., 2021). Similarly, Santos et al. described using a LAB consortium for fermenting a mixture of sheep, cow, and goat milks. This strategy resulted in the discovery of a new β -casein-derived bioactive polypeptide with significant antibacterial and antigelatinase properties (Santos et al., 2021). This polypeptide exhibited remarkable effectiveness against methicillin-resistant *S. aureus*, making it an ideal option for developing novel antibacterial nutraceuticals. Santos et al. (2021) presented all the casein-derived peptides obtained by in vitro fermentation.

19.2.1.3 Antimicrobial peptides found naturally in milk and dairy products

The three methods presented in the preceding sections demonstrated in vitro approaches for obtaining casein-derived AMPs. Several researchers, however, have investigated the direct nutritional value of milk and

dairy products by assessing their bioactive peptide content without further modification. Analyzing fresh milk and colostrum yielded the discovery of six different AMPs. Liu and Pischetsrieder, for example, found two potent bovine- α_{S2} -casein-derived AMPs in fresh bovine milk:

TKLTETEEKNRLNFKKISQRYQKFALPQYLYK and TVYQHQAAMKPWIPKTKVIPYVRYL (Liu and Pischetsrieder, 2018). These peptides were initially decrypted by the virtual screening method (Liu et al., 2015; Liu and Pischetsrieder, 2018). Furthermore, thus far, the peptide sequence TVYQHQAAMKPWIPKTKVIPYVRYL has been found in five different AMPs, indicating the presence of important antimicrobial motifs. Wang et al. also discovered the peptide YPVTQPLAPVHNPIIS, which was derived from the β -casein present in human breast milk (Wang et al., 2020). There was evidence of potent activity against *E. coli* and *Yersinia enterocolitica*. Moreover, the peptide demonstrated excellent performance in an experimental Neonatal necrotizing enterocolitis (NEC) mice model by reducing ileal mucosa damage. NEC is a deadly gastrointestinal disease that affects thousands of children in neonatal intensive care units. As a result, YPVTQPLAPVHNPIIS has been proposed as a potential candidate for developing treatment strategies for preventing and treating neonatal infections (Wang et al., 2020). Similarly, Fu et al. reported the discovery of LLNQELLNPTHQIYPV, a novel human β -casein-derived peptide (Fu et al., 2017). This peptide inhibited the pathogenic strains *E. coli*, *S. aureus*, and *Y. enterocolitica*. This peptide may also have the potential to help prevent infections in premature infants (Fu et al., 2017). These findings confirmed the critical role that human milk plays in the prevention and treatment of newborn gastrointestinal diseases.

On the other hand, several studies have found bioactive peptides in fermented dairy products. This is a direct result of the use of LAB strains in these products' manufacturing processes. As stated previously, LAB strains can release bioactive peptides from casein sources via the action of proteolytic enzymes. Kashk, a fermented dairy product popular in the Middle East, was recently studied by Folliero et al. to determine its antimicrobial and skin-repairing properties (Folliero et al., 2022). Several peptides, including the well-known Caseicin C, Caseicin 15, and Caseicin 17, were identified in the fraction with the most potent antibacterial activity. This fraction demonstrated remarkable activity against multiresistant clinical isolates of *S. aureus*, including multisensitive, methicillin-resistant, macrolide-resistant, quinolone-resistant, and β -lactamase producer strains. Furthermore, its low toxicity to HaCat (human keratinocytes) cells and its improved skin healing abilities make this fraction a promising research target for the treatment of

TABLE 19.2 The number of casein-derived antimicrobial peptides (AMPs) discovered by the different methods.

Release method	Enzyme	AMPs
Digestion	Pepsin	34
	Trypsin	4
	Chymotrypsin	2
Hydrolysis	Plasmin	11
	Chymosin	9
Mixture of proteases	NA	6
Fermentation	NA	10
Fresh milk/Colostrum	NA	6
Virtual screening	NA	15

antibiotic-resistant *S. aureus*-infected skin diseases (Folliero et al., 2022).

Similarly, Fialho et al. presented the extraction and identification of AMPs from the Canastra artisanal minas cheese (Fialho et al., 2018). Various AMPs were identified, including the α_{S1} -casein-derived peptides RPKHPIKHQ, RPKHPIKHQGLPQ, and RPKHPIKH QGLPQE. These fragments belong to Isracidine, a highly studied AMP also present in fresh bovine milk and colostrum (Fialho et al., 2018).

19.2.1.4 AMPs derived from casein sources

Several databases have reported the sequences and characteristics of milk-derived AMPs identified through various methods. Milk AMP (a database of antimicrobial milk peptides) and milk bioactive peptide database (MBPDB; a database of milk bioactive peptides) are two of the most popular and comprehensive sources (Nielsen et al., 2017). However, poor data quality, such as incorrect entries and entries from nonpeer-reviewed publications or non-in vitro validation, make it troublesome to use these databases for developing computation models, designing and implementing predictors, or performing macroanalysis (Nielsen et al., 2017). Based on this, MBPDB was chosen as the primary source for analyzing all previously reported casein-derived peptides. This database is the most reliable, up-to-date, and comprehensive source for searching milk-derived peptides or proteins with an individual in vitro confirmed bioactivity (MBPDB is available online at <http://mbpdb.nws.oregonstate.edu>) (Nielsen et al., 2017). All MBPDB entries for antimicrobial casein-derived peptides were carefully corroborated and confirmed for this chapter. Lists all previously reported casein-derived AMPs that have been individually tested in vitro and show the exact position of the motif in the casein source. These criteria

resulted in the identification of 95 peptides. The number of peptides discovered by each method is presented in Table 19.2. Furthermore, Table 19.3 categorizes peptides based on casein source and animal origin. The digestion with the pepsin method has contributed the most to the discovery of AMPs, yielding 34 novel peptides. Moreover, β -casein is the primary casein-derived AMPs source, with around 28 reported peptides. Furthermore, cow (*Bos taurus*) milk has been the most extensively researched source of AMPs, revealing 72 of the 95 confirmed casein-derived peptides with antimicrobial activity.

19.2.2 Computational approach

Peptide discovery and analysis is time-consuming, expensive, and labor-intensive. It requires in vitro screening of large libraries obtained through multiple mechanisms, such as those mentioned in the previous section. Over the last few decades, there has been a surge in the development of computational tools to speed up the process while lowering the associated time and costs and increasing its effectiveness (Ruiz Puentes et al., 2020). The primary goal of these methods is to evaluate the peptides' potential bioactivities, in this case, their antimicrobial capability. Machine-learning-based methods and molecular dynamics simulations are the most used methods (Ruiz Puentes et al., 2020). The former predicts whether an AMP will be present, whereas the latter investigates the potential mechanism of action against a specific type of membrane (Dziuba and Dziuba, 2014; Sansi et al., 2022). Both families of methods allow a reduction in the number of candidates to be evaluated in vitro, reducing the discovery process's time and costs. With the aid of these techniques, it is possible to accelerate the discovery of peptides by rationally exploring the vast

TABLE 19.3 Casein-derived antimicrobial peptides (AMPs) classified by casein source and animal origin.

Casein source	Species—origin	AMPs
α s1-casein	<i>Bos taurus</i>	18
	<i>Oryctolagus cuniculus</i>	1
	Total	19
α s2-casein	<i>B. taurus</i>	22
	<i>Ovis aries</i>	4
	Total	26
β -casein	<i>B. taurus</i>	16
	<i>Capra hircus</i>	7
	<i>Homo sapiens</i>	3
	<i>O. cuniculus</i>	2
	Total	28
κ -casein	<i>B. taurus</i>	16
	<i>C. hircus</i>	5
	<i>H. sapiens</i>	1
	Total	22

chemical space beyond the capability of experimental approaches (Ruiz Puentes et al., 2020).

The process of selecting AMP candidates using an in silico approach consists of three main steps: (1) building peptide libraries, (2) analyzing sequences, and (3) characterizing the most promising candidates. The obtained candidates should pass through an in vitro evaluation that allows confirming the peptides' bioactivity (Ruiz Puentes et al., 2022a). Fig. 19.2 shows the schematic representation of the in silico approach methods. In the case of casein-derived peptides, the library is typically obtained in three ways: in silico gastrointestinal digestion (Dziuba and Dziuba, 2014; Millan et al., 2022; Ryskaliyeva et al., 2019; Sansi et al., 2022), in vitro fermentation (as described in section 19.2.1.2), and identification of peptide sequences using reverse phase-high-performance liquid chromatography with subsequent mass spectrometry (Ashokbhai et al., 2022; Cirrincione et al., 2021; Iram et al., 2022; Pourjoula et al., 2020; Wei et al., 2022), or even from an already published database (Liu et al., 2015). The in silico gastrointestinal digestion can be performed in multiple software packages, including BIOPEP-UWM, PMAP, and PeptideCutter at ExPASy (Dziuba and Dziuba, 2014; Igarashi et al., 2009; Millan et al., 2022; Ryskaliyeva et al., 2019; Sansi et al., 2022).

Schematic representation of the computational approach for the discovery of casein-derived AMPs (created with BioRender.com). The process consists of two main steps: the first step is creating a peptides library,

which can be obtained by an in vitro or an in silico technique. The second corresponds to the library analysis by a machine-learning algorithm, followed by a selection of the best candidates based on physicochemical properties. Finally, a molecular docking simulation can be performed to understand the action mechanism of the peptide.

BIOPEP-UWM software enables protein digestion by up to three enzymes simultaneously (Minkiewicz et al., 2019). Briefly, under the "BIOPEP-UWM" tab of the website (<https://biochemia.uwm.edu.pl/en/biopep-uwm-2/>), the following steps are Bioactive peptides → Analysis → Enzyme(s) action → For your sequence. After entering the desired sequence, up to three enzymes from a list of 33 options can be selected for digestion. The most commonly used digestion strategies include a cocktail of pepsin, trypsin, and chymotrypsin A or a separate analysis of the same sequence with multiple enzymes (Dziuba and Dziuba, 2014; Millan et al., 2022; Sansi et al., 2022). Regarding PeptideCutter (Gasteiger et al., 2005), it enables the digestion of the sequences by up to 28 enzymes and some other chemical compounds irrelevant for gastrointestinal digestion. This software, interestingly, returns not only the cleavage sites but also the probabilities of each cut. It also computes the cleavage analysis with pepsin under two pH conditions and chymotrypsin under two specificity conditions. Both enzymes are essential in the in vivo process of milk gastrointestinal digestion. The probabilities returned and the option of multiple conditions for an enzyme give researchers more control over the simulated digestion process, which translates into a

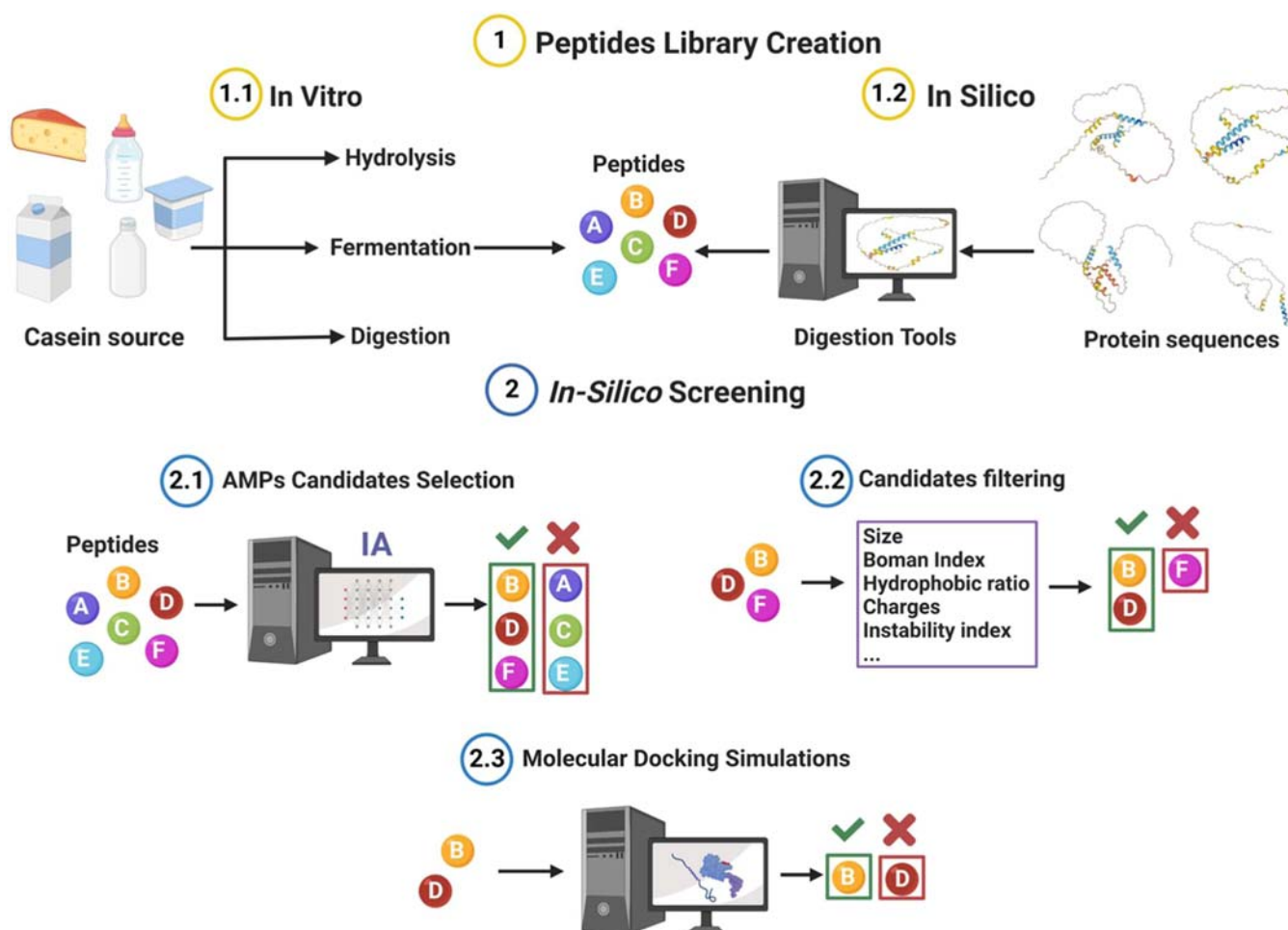


FIGURE 19.2 Schematic representation of the computational approach for discovering casein-derived AMPs.

more robust peptide selection for the library (Gasteiger et al., 2005). Finally, even though multiple works have used PMAP in the past, the server at the reported URL is no longer active as of 2023 (Igarashi et al., 2009).

After establishing a peptide library, researchers use online public servers to analyze peptide sequences and determine the likelihood of their antimicrobial capability. The most used server is CAMPR3, which includes four machine-learning models for analyzing sequences: a support vector machine (SVM), a random forest (RF), a discriminant analysis (DA), and artificial neural networks (ANNs) (Waghu et al., 2016). All the models were implemented in the software R®. Each method classifies the sequence using 64 features related to AA composition, solvent accessibility, helical propensity, secondary structure distribution, and multiple physicochemical indexes correlated to charge, hydrophobicity, polarity, and hydrophilicity. The SVM was built with “Kernlab,” which employs a polynomial kernel with degrees, scale, and offset set to 3, 0.01, and 1, respectively. The RF was built with the “randomForest” package and can contain up to

500 decision trees. The ANN was built with the “nnet” (neural networks) package, but the authors failed to provide further details about the implementation. Finally, the DA was built with the “SPSS” package, which included a stepwise selection algorithm with backward elimination (Waghu et al., 2016). In November 2022, a new server version, CAMPR4, was released. It no longer employs DA in its methods but instead employs different models to analyze natural and synthetic peptides to improve the estimation of each category (Gawde et al., 2023).

In addition, the “Profile of potential biological activity” function of the BIOPEP-UWM database is another server that can be used to find potential antimicrobial segments within a specific peptide (Minkiewicz et al., 2019). Similarly, for the “Enzyme action” function, the following steps are Bioactive peptides → Analysis → Profile of potential biological activity → For your sequence. The sequence can be tested against all or just one bioactivity in the database. In this case, the relevant bioactivities are antibacterial, antifungal, and antiviral. In summary, the software compares the sequence of interest to all peptides

reporting the desired bioactivity. It returns all motifs within the peptide reported to have the desired bioactivity (Minkiewicz et al., 2019). Researchers can use this database report as a proxy for antimicrobial capability. Similarly, the MBPDB allows the user to search for the bioactivity of each sequence by comparing it to the database's reported peptides via a similarity score threshold (Nielsen et al., 2017).

Finally, the antimicrobial sequence scanning system server has been used to identify AMPs from casein (Tomazou et al., 2019; Torrent et al., 2012). This method differs from the others because it does not require a peptide library; instead, it analyzes the entire protein and looks for antimicrobial motifs. These regions are discovered using a bactericidal propensity index value (PIV) for each AA. The PVs for all the AAs were calculated based on the experimental research of Torrent et al. In a nutshell, sections of 12 AAs within the protein and up to two gaps are analyzed to calculate the overall average PV. Segments are classified as antimicrobial or not based on a predefined threshold (Torrent et al., 2012, 2009).

Molecular docking simulations are also used to gain a better understanding of the antimicrobial mechanisms of the candidates. This method enables a more detailed understanding of the atomic scale interaction between a small molecule and a protein. This method characterizes peptide behavior based on the protein binding site and elucidates fundamental biochemical processes. The docking procedure consists of two basic steps: predicting the ligand conformation and position and determining the orientation within the binding site and the corresponding binding affinity. These simulations have shown excellent performance in locating information at the atomic level, which would be extremely difficult to obtain experimentally (Meng et al., 2011). Molecular docking has also been used to investigate the interaction of peptides with the binding sites of some essential proteins involved in antimicrobial activity, such as dihydropteroate synthase (Sansi et al., 2022), MurD ligase (Iram et al., 2022), hydroxymethylglutaryl-CoA synthase (Ashokbhai et al., 2022), and dihydrofolate reductase (Ashokbhai et al., 2022). The primary goal is to guide experimental research by understanding the interactions between the peptides and the proteins involved.

After any of the aforementioned computational tools has identified some AMPs candidates, a characterization based on some physicochemical properties is typically performed through comparisons with known AMP properties (Dziuba and Dziuba, 2014; Iram et al., 2022; Prabakusuma and Aleryani, 2022; Ruiz Puentes et al., 2022a; Sansi et al., 2022; Tomazou et al., 2019). The following are some of the most commonly used properties:

1. Sequence length. Known AMPs are usually small (<100 AA) and, on average, are between 4 and 30 AA (Dziuba and Dziuba, 2014; Ruiz Puentes et al., 2022a). However, some researchers may select smaller sizes depending on the computational model's capability and available resources.
2. Molecular weight. No specific range of molecular weight for AMPs has been reported.
3. Isoelectric point. It has been reported to be around 10 as it correlates well with the disruption-membrane capability of most AMPs, which is similar to that of soap and detergents (Moretta et al., 2020; Osorio et al., 2015; Torrent et al., 2011).
4. Net charge at pH 7.0. Natural AMPs is mostly cationic or neutral to interact with the negatively charged bacterial membranes (Torres et al., 2019).
5. Hydrophobic ratio. In naturally occurring peptides, it usually varies between 0.4 and 0.6. Hydrophobicity is a fundamental factor, given the high percentage of lipids in bacterial membranes (Torres et al., 2019).
6. Aliphatic index. It is usually high (>40), which is required for superior heat stability (Li et al., 2016).
7. Instability index. It is usually below or around 40 to ensure a high bioavailability and adequate half-life (Dziuba and Dziuba, 2014; Ruiz Puentes et al., 2022b).
8. Bowman index. In naturally occurring peptides, this index varies approximately between -7 and 7 and estimates the potential of peptides to interact with proteins. Low values suggest a high antimicrobial activity without promoting significant undesirable side effects. Higher values indicate multifunctional roles within the cell, leading to a higher probability of undesirable side effects (Azad et al., 2011; Nam et al., 2014; Ruiz Puentes et al., 2022a).

Other characteristics that might be evaluated include the amphipathicity (which can correlate with the structure and mechanism of action), the helical structure, the toxicity and allergenicity for humans, and their GRAVY index and solubility in water, which is fundamental for posterior in vitro or in vivo experimentation (Dziuba and Dziuba, 2014; Iram et al., 2022; Prabakusuma and Aleryani, 2022; Sansi et al., 2022; Tomazou et al., 2019). Multiple software packages and libraries exist to compute all the mentioned properties: Pep-Calc (Lear and Cobb, 2016), ProtParam (Gasteiger et al., 2005), APD3 algorithm: Antimicrobial Peptide Calculator and Predictor (Wang et al., 2016), ToxinPred (Gupta et al., 2013), AllergenFP (Dimitrov et al., 2014), CellPPD (Gautam et al., 2013), HeliQuest (Gautier et al., 2008), PepFun (Ochoa and Cossio, 2021), Modlamp (Müller et al., 2017), CAMPR4 (Gawde et al., 2023), and Peptides both in Python and R® (Osorio et al., 2015).

Table 19.4 presents multiple sequences obtained through a combination of computational tools that have been predicted as promising AMPs candidates.

TABLE 19.4 Casein-derived peptides predicted by computational tools: promising antimicrobial peptides (AMPs) candidates.

Casein source	AMP sequence	Type of digestion	Method of analysis	References
αs1-Casein	KKYKVPQL	In silico BIOPEP	CAMP	Dziuba and Dziuba (2014)
αs2-Casein	DDKHYQKA			
β-Casein	EMPFPK			
β-Casein	ITRINKKIEKFQS			
β-Casein	ITRINKKIEKF			
κ-Casein	QEQNQEQP	In silico BIOPEP	CAMPR3	Sansi et al. (2022)
αs1-Casein	SGK			
αs1-Casein	IQK			
αs2-Casein	SIR			
αs2-Casein	AIH			
αs2-Casein	TQPK			
β-Casein	GPVR			
β-Casein	AVPQR			
β-Casein	AIAR			
β-Casein	GVPK			
β-Casein	SQPK			
β-Casein	PVPQK			
β-Casein	IH			
β-Casein	VPK			
αs1-Casein	D. APSFSDIPNPIGSENSGKT. T			
αs1-Casein	D. APSFSDIPNPIGSEN. S			
αs1-Casein	D. IPNPIGSENSGKTTMPL. W			
αs1-Casein	D. APSFSDIPNPIGSENSGK. T	Sequence comparison to milkAMP database	Sequence comparison to milkAMP database	Folliero et al. (2022)
αs1-Casein	D. APSFSDIPNPIGSE. N			
αs1-Casein	D. APSFSDIPNPIGSENSG. K			
αs1-Casein	Q. GLSPEVPNENLLR. F			
αs1-Casein	D. IPNPIGSENSGKITMPLW			
αs1-Casein	D. IPNPIGSENSGKTTM(+15.99)PLW			
αs1-Casein	N. SGKTTMPLW			
αs1-Casein	F. VVAPFPEVFR. K			
αs1-Casein	D. APSFSDIPNPIGSE. N			
αs1-Casein	N. PIGSENSGKTTMPLW			
αs1-Casein	D. IPNPIGSENSGKTTMPLW			
αs1-Casein	D. APSFSDIPNPIGSENSGKTTMPLW			
αs1-Casein	D. APSFSDIPNPIGSENSGKITMPLW			
β-Casein	L. YQEPVLGPVRGPFPI. I			
β-Casein	L. YQEPVLGPVRGPFPI. L			

(Continued)

TABLE 19.4 (Continued)

Casein source	AMP sequence	Type of digestion	Method of analysis	References
β -Casein	L. LYQEPVLGPVRGPFPIV			
β -Casein	L. YQEPVLGPVRGPFPIV			
β -Casein	L. YQEPVLGPVRGPFPIV. V			
β -Casein	E. PVLGPVRGPFPIV			
β -Casein	L. GPVRGPFPIV. V			
β -Casein	L. GPVRGPFPIV			
β -Casein	G. PVRGPFPIV			
κ -Casein	F. LPYPYAKPV. A			
κ -Casein	F. LPYPYAKPIA. V			
κ -Casein	Q. FLYPYAKPV. A			
κ -Casein	N. NQFLYPYAKPV. A			
κ -Casein	F. LPYPYAKPI. A			
α s2-Casein	TVYQYQK	In silico PeptideCutter	BIOPEP and AMP Scanner V2	Ryskaliyeva et al. (2019)
α s2-Casein	PWNHIKRYF			
α s2-Casein	LKKISORYOKFALPOY			
α s2-Casein	PWIOPKTKVIPYVRYL			
α s1-Casein	HQGLPQE	In vitro digestion	BIOPEP, EROP- MOSCOW, and MBPDB	Wei et al. (2022)
α s1-Casein	LNENLLR			
α s1-Casein	FS			
α s1-Casein	DIPNPIGSENSEK			
α s2-Casein	NPWDQVKR			
α s2-Casein	VYOHOKAMKPWIOPKTKVIPY			
β -Casein	MPFPKY			
κ -Casein	FSDKIAK			
κ -Casein	VQVTSTAV			
β -Casein	DPATQPIVPVHNIP	In vitro fermentation	Comparison against MBPD database	Cirincione et al. (2021)
β -Casein	FDPATQPIVPVHNIPV			
β -Casein	PATQPIVPVHNIPV			
β -Casein	PATQPIVPVHNIPVI			
β -Casein	PATQPIVPVHNIPVIV			
β -Casein	QPVPYPQRDTPVQAF			
β -Casein	SKVAPFPQPVVPYPQ			
β -Casein	VAPFPQPVVPYPQR			
β -Casein	VAPFPQPVVPYPQ			

19.2.2.1 Novel approaches: deep learning–based strategies

Although traditional machine-learning methods were previously used to discover potential AMPs, newer and

better deep learning–based algorithms have been designed in the last decade ([Ruiz Puentes et al., 2022a](#)). Furthermore, in the last few years, those methods have been used in general pharmaceuticals discovery as well

as in the specific case of AMPs discovery from multiple sources, with outstanding results (Ruiz Puentes et al., 2022a, 2022b; Stokes et al., 2020). Some of the methods used are based on natural language processing algorithms, assuming that, as with the problem of natural language understanding, the most important information is contained in the sequential order of their elements (Veltri et al., 2018; Wu et al., 2019; Zohora et al., 2019). Algorithms based on a graph representation of the molecules have also attracted significant attention, which is ideal given the non-Euclidean nature of chemical data (Ruiz Puentes et al., 2022a). In this case, molecular graph representations allow for more accurate and explicit modeling of atom and bond spatial configurations. These methods provide a global understanding of the structure of molecules via a message-passing algorithm that allows information to flow between all atoms and their bonds (Stokes et al., 2020; Yang et al., 2019). In both cases, the goal is not to use precomputed features but rather to allow networks to learn about the chemical space based on the data gathered thus far for AMPs. In this regard, it is possible to train substantially large networks capable of learning detailed information about the underlying chemical space (Li et al., 2022a; Lin et al., 2021; Ruiz Puentes et al., 2022a; Veltri et al., 2018).

Furthermore, because they do not rely on hand-crafted features but instead learn on their own, they may be able to discover AMPs outside the data distribution of known AMPs, as they have been the case with antibiotics (Stokes et al., 2020). As a result, these methods have emerged as suitable and potent alternatives to traditional machine learning approaches based on their superior performance. This property will lead to the discovery of novel casein-derived AMPs with remarkable antimicrobial activity, thereby leading to significant time and cost savings.

19.3 Conclusions

AMPs have prompted the interest of researchers as promising platforms for developing novel food preservatives, nutraceuticals, functional foods, and next-generation drugs. Fresh milk, colostrum, and dairy products have shown enormous potential as AMPs sources. Casein, the main milk-derived protein, has been widely reported as an important source of peptides with potent antibacterial and antifungal activity. Furthermore, various peptides have demonstrated remarkable activity against pathogenic strains of international concern, such as multidrug-resistant *S. aureus*, *L. monocytogenes*, and *C. albicans*. However, the potential and safety of casein-derived AMPs have yet to be fully assessed. The lack of studies examining the performance and impact of casein-derived AMPs in

complex environments such as animal models makes it difficult to determine their true potential. Furthermore, the high costs of isolating, evaluating, and selecting the best candidates using traditional methods have significantly slowed the progress of this field. As a result, only a few peptide candidates make it to the clinical level or reach advanced stages in the food industry. Alternatively, novel strategies, such as computational approaches, have gained significant attention as versatile tools for discovering and predicting the function of AMPs at a much lower cost. Nonetheless, current approaches employ antiquated strategies that have shown suboptimal performance as peptide predictors. These strategies, primarily based on feature analysis, are imprecise, and the results provide only limited progress in discovering AMPs. Conversely we propose implementing powerful computational tools such as deep learning-based strategies with graph representations as a more appropriate alternative for accelerating the discovery pipeline. This combination allows learning the chemical structure, a critical feature that has been successfully used for pharmaceutical analysis with remarkable results. Finally, there is an urgent need to assess the mechanisms of action and biosafety of peptides in more complex environments. This will enable the identification of viable candidates for developing novel food preservatives and pharmaceutical formulations. Furthermore, by utilizing deep learning-based strategies to accelerate the selection of the best candidates, time and costs are expected to be reduced within the next decade.

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CASEIN

Structural Properties, Uses, Health Benefits and Nutraceutical Applications

Edited by
Mamdouh El-Bakry
Bhavbhuti M. Mehta

An exploration of casein, its types, properties, and applications in the food and pharmaceutical industries

Casein: Structural Properties, Uses, Health Benefits and Nutraceutical Applications investigates casein properties, uses, and applications in food and non-food products, in addition to exploring its health benefits and uses in manufacturing, such as in cheese products, along with an in-depth discussion on the future scope, challenges, and market trends of this protein.

Casein: Structural Properties, Uses, Health Benefits and Nutraceutical Applications is an excellent reference for food scientists, dairy researchers, pharmaceutical scientists, students and researchers studying related fields.

Key Features

- Provides comprehensive coverage of casein, the main milk protein that has many applications and uses
- Includes suggested reading for further information
- Addresses a wide-range of related topics, including non-food applications of casein

Mamdouh El-Bakry, Ph.D., Regulatory Affairs Technologist, Food Supplements and Natural Cosmetics, Cartagena, Spain

Bhavbhuti M. Mehta, Ph.D., Associate Professor and Head, Dairy Chemistry Department, Sheth M.C. College of Dairy Science, Kamdhenu University, Anand, Gujarat, India



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